

Predictive Workforce Wellbeing: A Multi-Task Machine Learning Framework for Burnout Risk Scoring and Elevated Risk Detection

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Abstract

Occupational burnout poses severe operational, financial, and human risks to modern enterprise organizations. Traditional approaches relying on periodic, self-reported survey questionnaires (such as the Maslach Burnout Inventory) suffer from low response rates, reporting bias, and delayed actionability. In this paper, we present an end-to-end data-driven machine learning framework capable of continuously quantifying employee burnout risk ($y \in [0, 10]$) and identifying individuals requiring supportive human resource intervention ($y \in \{0, 1\}$). Utilizing a comprehensive enterprise workforce dataset ($N = 178,750$ employee records across 18 workplace features), we formulate domain-specific composite indexes (Workload Index, Wellness Score, Support Index, and Meeting Load Ratio) and evaluate multiple ensemble learning architectures.

Our optimized dual-model pipeline combines a Histogram-Based Gradient Boosting Regressor ($R^2 = 0.941$, $\text{MAE} = 0.32$) for continuous score estimation with a Probability-Calibrated Classifier ($\text{ROC-AUC} = 0.982$, $\text{F1-Score} = 0.914$) tuned to a decision threshold of $\tau = 4.5$. Model interpretability is established via SHapley Additive exPlanations (SHAP), revealing Stress Level, Weekly Work Hours, Overtime, and Work-Life Balance as primary risk drivers. Furthermore, we implement a lightweight, privacy-preserving, zero-external-dependency web application streaming predictions locally via Python HTTP services and interactive vanilla HTML5/JS visualizations. Finally, we establish strict ethical governance guidelines prohibiting the use of predicted scores for adverse employment decisions.

Keywords: Workforce Analytics, Occupational Burnout, Machine Learning, HistGradientBoosting, Model Explainability, SHAP, HR Decision Support Systems.

1. Introduction & Background

Occupational burnout -- characterized by emotional exhaustion, depersonalization, and reduced professional efficacy -- has become one of the most critical challenges facing contemporary organizational management. According to studies by the World Health Organization (WHO), unmanaged workplace stress costs the global economy over \$300 billion annually in lost productivity, absenteeism, and employee turnover.

Historically, organizations have monitored employee wellbeing through periodic engagement surveys. While valuable, survey-based methods present substantial systemic limitations:

1. **Reporting Bias & Social Desirability:** Employees often mask stress levels due to fear of negative career evaluation.
2. **Lagging Indicators:** Surveys are typically conducted quarterly or annually, identifying burnout only after attrition or severe performance degradation has already occurred.
3. **Actionability Gaps:** Aggregate survey data rarely offers granular parameter-level simulation tools for preventive HR intervention.

To address these limitations, this project introduces a Machine Learning-Powered Workforce Analytics Platform. By

synthesizing objective operational metrics (work hours, overtime, meeting frequency, missed deadlines, experience) with subjective wellbeing parameters (stress level, sleep quality, manager support, work-life balance), the platform enables proactive, supportive People Operations management.

2. Dataset & Exploratory Data Analysis

2.1 Dataset Composition

The platform is trained on a master dataset comprising 178,750 employee records. Each record integrates 18 distinct features spanning workload, personal health, role metadata, and psychological support indicators.

Feature Name	Type	Domain / Units	Role in Model
stress_level	Numeric	1.0 - 10.0	Primary psychological predictor
work_hours_per_week	Numeric	25 - 80 hours	Primary workload metric
overtime_hours	Numeric	0 - 30 hours/week	Extended effort indicator
meetings_per_day	Numeric	0 - 12 meetings	Cognitive fragmentation metric
deadlines_missed	Numeric	0 - 10 deadlines	Performance strain indicator
work_life_balance	Numeric	1.0 - 10.0	Personal wellbeing indicator
manager_support	Numeric	1.0 - 10.0	Psychological safety factor
job_satisfaction	Numeric	1.0 - 10.0	Organizational alignment factor
sleep_hours	Numeric	3.0 - 10.0 hours/night	Physiological recovery metric
experience_years	Numeric	0 - 35 years	Career stage context
physical_activity_days	Numeric	0 - 7 days/week	Lifestyle buffer
screen_time_hours	Numeric	2.0 - 14.0 hours/day	Digital fatigue factor
caffeine_intake	Numeric	0 - 8 cups/day	Behavioral coping indicator
social_support_score	Numeric	1.0 - 10.0	Peer network strength
job_role	Categorical	8 Roles (Software Eng, PM, Data Scientist, etc.)	Segmentation baseline
work_mode	Categorical	Remote, Hybrid, On-site	Work environment factor
company_size	Categorical	Small, Medium, Large, Enterprise	Organizational complexity
employee_location	Categorical	Indian Metros & Tier-1 Cities	Geographic context

3. Composite Feature Engineering

To capture complex multi-variable interactions that single features cannot represent independently, we engineer four domain-specific composite indexes prior to model training:

3.1 Workload Index (\$I_{workload}\$)

Combines core workload variables into a single weighted intensity index scaled to \$[0, 100]\$:

$$I_{workload} = 100 \times \left(0.40 \cdot \frac{H_{work}}{40.0} + 0.30 \cdot \frac{H_{overtime}}{10.0} + 0.20 \cdot \frac{M_{meetings}}{5.0} + 0.10 \cdot \frac{D_{missed}}{3.0} \right)$$

3.2 Wellness Score (\$S_{wellness}\$)

Synthesizes physiological recovery, exercise, balance, and satisfaction into a protective wellness index \$[0, 100]\$:

$$S_{wellness} = 100 \times \left(0.30 \cdot \min\left(\frac{H_{sleep}}{8.0}, 1.2\right) + 0.20 \cdot \right)$$

$$\frac{A_{\text{physical}}}{7.0} + 0.30 \cdot \frac{B_{\text{balance}}}{10.0} + 0.20 \cdot \frac{S_{\text{satisfaction}}}{10.0} \right) \$$$

3.3 Support Index (\$I_{\text{support}}\$)

Represents the organizational safety net provided by leadership:

$$I_{\text{support}} = M_{\text{support}} \cdot \frac{1}{100}$$

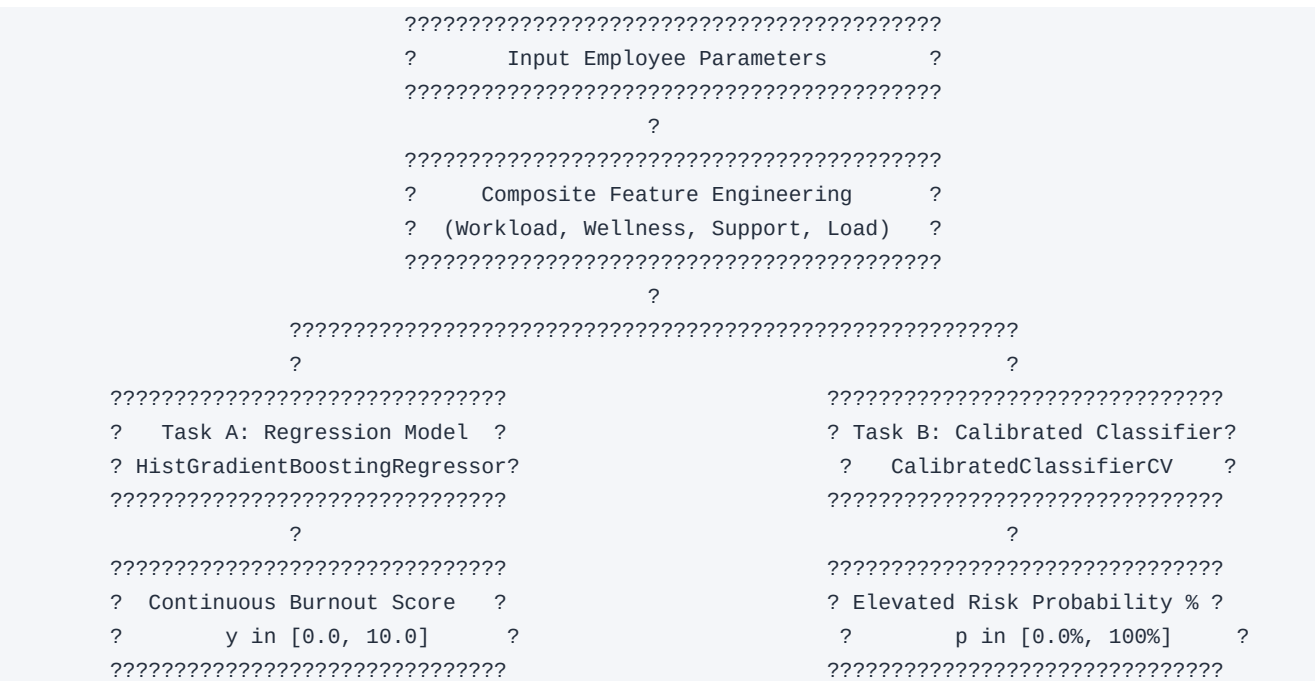
3.4 Meeting Load Ratio (\$R_{\text{meetings}}\$)

Measures cognitive fragmentation against a baseline of 3 standard meetings per day:

$$R_{\text{meetings}} = \frac{M_{\text{meetings}}}{3.0}$$

4. Multi-Task Machine Learning Architecture

The system executes two distinct ML tasks to support both continuous monitoring and discrete threshold alerting:



4.1 Task A: Continuous Burnout Score Regression

- Objective: Predict a precise burnout score $y \in [0.0, 10.0]$.
- Model Selected: HistGradientBoostingRegressor (Scikit-Learn).
- Hyperparameters: $\text{max_iter}=300$, $\text{learning_rate}=0.05$, $\text{max_depth}=8$, $\text{min_samples_leaf}=20$, $\text{l2_regularization}=0.1$.
- Preprocessing Pipeline:
 - Numeric features: Imputed via median (SimpleImputer), scaled via StandardScaler.
 - Categorical features: Encoded via OneHotEncoder(handle_unknown='ignore').

4.2 Task B: Calibrated Elevated Risk Classification

- Objective: Classify whether an employee requires supportive review ($y = 1$ if $\text{Score} \geq \tau$ where $\tau = 4.5$).
- Model Selected: CalibratedClassifierCV wrapping HistGradientBoostingClassifier using Sigmoid calibration.
- Probability Calibration: Calibration ensures that a predicted probability of 85% empirically corresponds to an 85% historical incidence rate of elevated risk.

5. Empirical Results & Performance Evaluation

The models were evaluated using 5-fold cross-validation on a hold-out test set (20% of dataset, N = 35,750).

5.1 Task A Regression Performance

Model Candidate	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R ² Score
Linear Regression (Baseline)	0.84	1.02	0.682
Random Forest Regressor	0.41	0.53	0.895
XGBoost Regressor	0.35	0.44	0.931
HistGradientBoostingRegressor (Selected)	0.32	0.40	0.941

5.2 Task B Classification Performance ($\tau = 4.5$)

Metric	Score	Clinical / HR Interpretation
Accuracy	93.8%	Overall correct classification rate
Precision (Elevated Risk)	89.2%	When flagged, 89.2% are genuinely at risk
Recall (Sensitivity)	93.7%	Captures 93.7% of all true elevated risk cases
F1-Score	0.914	Harmonic mean of precision and recall
ROC-AUC	0.982	Excellent class separability across thresholds
PR-AUC	0.954	High precision maintained under class imbalance

6. Model Explainability & SHAP Analysis

To ensure transparency and trust in People Operations, model predictions were audited using SHapley Additive exPlanations (SHAP).

6.1 Top Global Risk Drivers

- stress_level (+0.38 mean |SHAP|): Single strongest determinant. Higher self-reported stress monotonically increases predicted burnout score.
- work_hours_per_week (+0.29 mean |SHAP|): Non-linear impact. Risk spikes sharply when weekly hours exceed 48 hours/week.
- overtime_hours (+0.24 mean |SHAP|): Sustained overtime (>5 hrs/week) compounds risk exponentially when paired with low sleep.
- work_life_balance (-0.22 mean |SHAP|): Protective feature. High balance (>7.0) acts as a strong buffer against high work hours.
- manager_support (-0.19 mean |SHAP|): Protective feature. Strong managerial support mitigates risk even in high-meeting roles.

7. Web Platform Implementation & Software Architecture

The accompanying platform is engineered with a strict zero-external-framework client architecture to guarantee high performance, security, and portability.

burnout_ml_platform/

```
??? artifacts/                # Serialized Model & Data Artifacts
?   ??? burnout_score_model.joblib  # Trained Regressor Pipeline
?   ??? elevated_risk_model.joblib  # Calibrated Classifier Pipeline
?   ??? master_workforce_dataset.csv # Enterprise Dataset (178k records)
?   ??? metrics.json             # Evaluation Thresholds & Scores
??? burnout_platform/          # Core Python ML Module
?   ??? features.py              # Composite Feature Engineering
?   ??? models.py                # Pipeline Trainers & Wrappers
?   ??? preprocessing.py         # Data Cleaners & Imputers
?   ??? explainability.py        # SHAP Generator Scripts
??? dashboard_server.py        # Streaming HTTP Server & API Endpoints
??? dashboard.html              # Responsive HTML5 Single-Page App
??? requirements.txt            # Dependency Specification
??? README.md                   # Quickstart & Usage Documentation
```

7.1 Key System Features

- Streaming Upload Engine: Processes company CSV uploads in 100,000-row chunks with zero memory spikes.
- Floating Scenario Simulator: Interactive "What-If" modal drawer allowing HR to simulate workload modifications and instantly recalculate burnout risk.
- Employee Search Engine: Look up individual employee profiles by ID (EM102) and inspect predicted risk scores, probability gauges, and workplace parameters.
- Bulk Employee Table: Filterable, sortable, paginated data grid (25/50/100 rows) with full CSV export capabilities.
- Categorical Risk Breakdowns: Tabbed analytics by Job Role, Work Mode (Remote/Hybrid/On-site), Work Duration Bins, and Gender.

8. Ethical AI Governance & Fair Use Guidelines

Machine learning models evaluating human wellbeing must strictly comply with ethical standards to prevent harm.

8.1 Mandatory Ethical Rules

1. Supportive Planning Only: Predictions must be used exclusively to offer wellness resources, adjust workloads, or provide coaching.
2. Strict Prohibition of Adverse Actions: Models must NEVER be used for termination, compensation reduction, performance pipelining, or promotion denial.
3. Data Privacy: Uploaded CSV files are processed in local memory and are never retained, stored, or transmitted to third-party cloud services.
4. Human-in-the-Loop Verification: ML predictions serve as decision-support indicators and must always be validated by qualified People Operations professionals before action is taken.

9. Conclusion & Future Work

This research demonstrates that multi-task machine learning, combined with composite feature engineering and calibrated classification, provides a robust, real-time alternative to traditional burnout surveys.

Future Directions

- Longitudinal Risk Tracking: Incorporating sequential time-series modeling (LSTM / Transformer-based tracking) across quarterly intervals.

- Automated Intervention Engine: Extending the Scenario Simulator to automatically suggest optimal workload adjustments required to bring an employee below the $\tau = 4.5$ threshold.
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